



DEPARTMENT OF INFORMATION TECHNOLOGY

CS23621- Mobile Application Development Laboratory

MINI-PROJECT REPORT

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ColorVis

Color Blindness Helper Android Application

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1. Introduction

ColorVis is an Android application designed to assist individuals with color blindness by providing color detection and simulation functionalities. The application allows users to upload images, simulate different types of color blindness, and identify colors through tap-based color detection. The app focuses on accessibility, readability, and ease of use.

2. Objective

- To help users understand how images appear under different color blindness conditions.
- To provide accurate color information in textual format.
- To improve accessibility awareness through guidance and tips.
- To build a simple offline Android application using Kotlin and XML.

3. Problem Statement

People with color vision deficiency often face difficulties in distinguishing certain colors in daily life, educational materials, websites, and digital applications. Many existing solutions require internet access or complex tools. A lightweight offline Android solution can improve accessibility.

4. Existing System

Existing applications mainly focus on online services or complex AI-based systems. Some applications do not provide proper textual color information or accessibility guidance.

5. Proposed System

ColorVis provides a simple and efficient Android-based solution for color blindness assistance. The application uses bitmap image processing to simulate Protanopia, Deuteranopia, and Tritanopia locally without requiring internet connectivity.

6. Features of the Application

- Splash screen with app branding
- Image upload from gallery
- Color blindness simulation
- Tap-based color detection
- RGB and HEX color display
- Accessibility tips section
- Dark mode support
- Offline functionality

7. Use Case

Users can upload an image and analyze how it appears for different types of color blindness. The user can tap any part of the image to identify the exact color name, RGB values, and HEX code. The app can also be used for accessibility learning and UI design improvements.

8. Methodology

- Bitmap image processing was used for image manipulation.
- Pixel-level color extraction was used for color detection.
- Matrix-based RGB transformations were used for simulation filters.
- Material Design components were used for user interface development.

9. Technologies Used

Technology	Purpose
Kotlin	Application Development
XML Layouts	UI Design
Android SDK	Android Framework
Material Design	Modern UI Components
Bitmap Processing	Image Processing
Android Studio Panda	Development Environment

10. System Requirements

- Android Studio Panda Version
- Minimum SDK: 24
- Android Device or Emulator
- Windows/Linux/macOS System
- 4GB RAM or higher

11. Modules Description

Splash Screen: Displays application logo and branding.

Home Screen: Provides navigation to all major features.

Image Upload Module: Allows users to select images from gallery.

Simulation Module: Applies color blindness filters to uploaded images.

Color Detection Module: Detects and displays tapped pixel color information.

Accessibility Tips Module: Displays accessibility recommendations.

12. Advantages

- Works completely offline
- Simple and beginner-friendly
- Improves accessibility awareness
- Lightweight application
- Easy to use interface

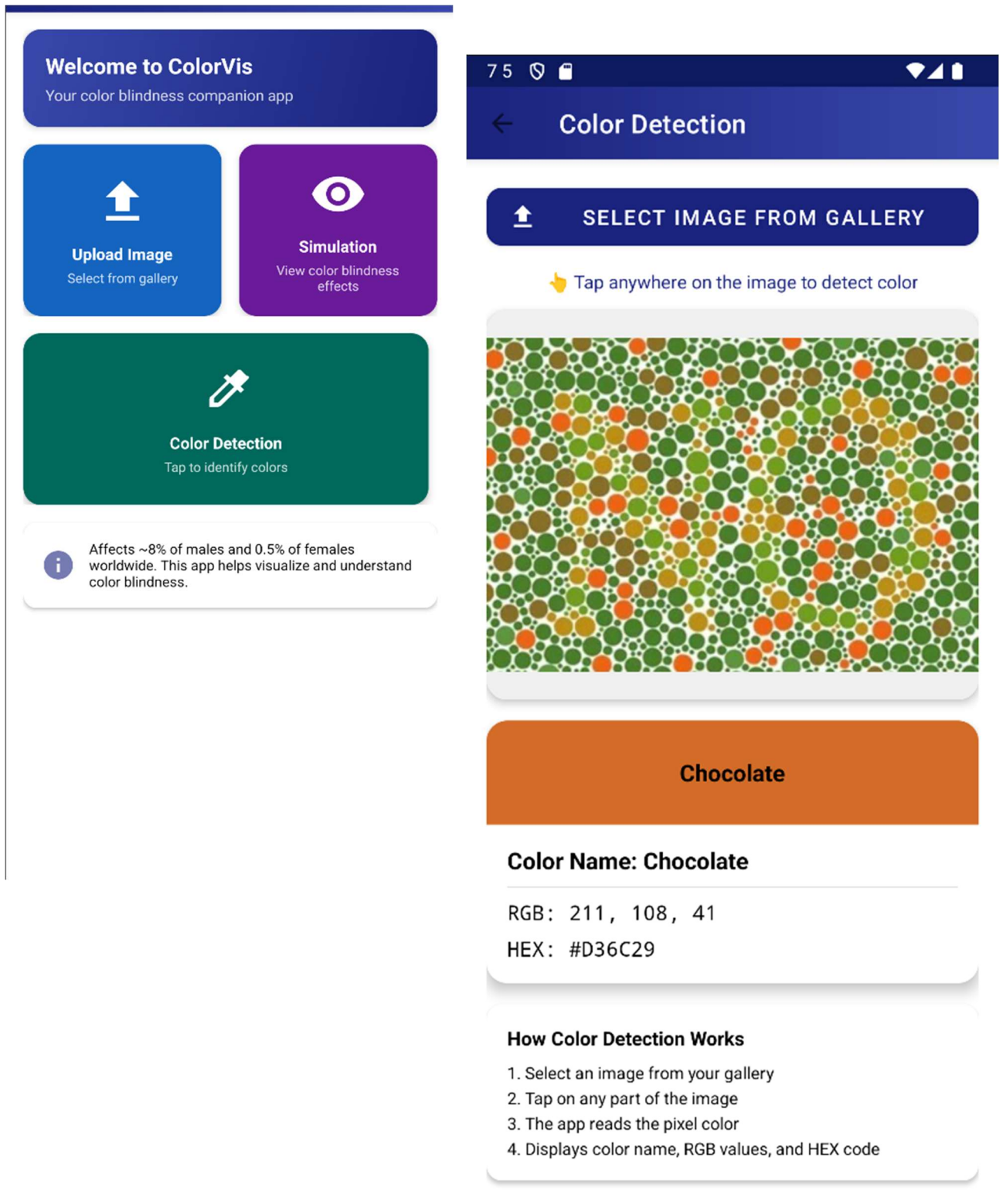
13. Limitations

- Limited to static image analysis
- No real-time camera processing
- Basic color naming support

14. Future Enhancements

- Real-time camera support
- Advanced accessibility analytics
- Additional color correction modes
- Voice assistance integration

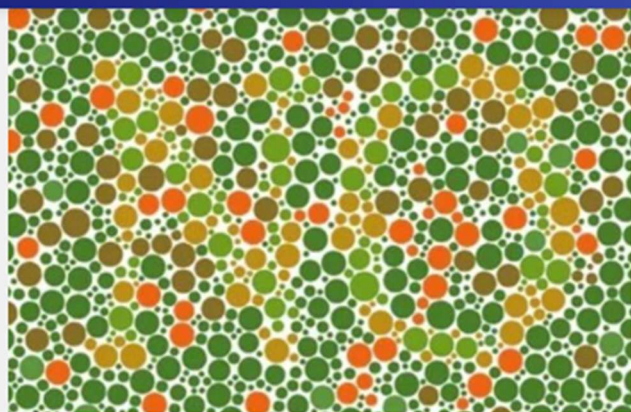
15. Result



7 58



Color Blindness Simulation



Choose a simulation type:

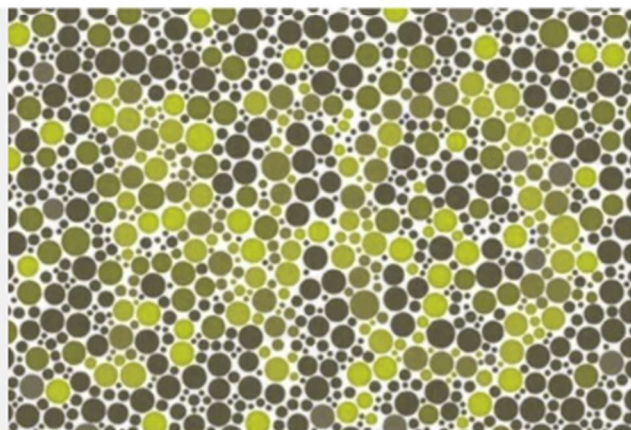
PROTANOPIA (RED BLIND)

DEUTERANOPIA (GREEN BLIND)

TRITANOPIA (BLUE BLIND)

SHOW ORIGINAL

Deuteranopia Simulation



16. Conclusion

ColorVis successfully demonstrates an accessible Android application that assists users in understanding color blindness effects and identifying colors accurately. The application provides a clean interface, offline support, and practical accessibility features.